

Overview of Information retrieval

Subject: Information retrieval

1.

Overview An information retrieval process begins when a user enters a query into the system. Queries are formal statements of information needs, for example search strings in web search engines. In information retrieval, a query does not uniquely identify a single object in the collection. Instead, several objects may match the query, perhaps with different degrees of relevance. An object is an entity that is represented by information in a content collection or database. User queries are matched against the database information. However, as opposed to classical SQL queries of a database, in information retrieval the results returned may or may not match the query, so results are typically ranked. This ranking of results is a key difference of information retrieval searching compared to database searching. Depending on the application the data objects may be, for example, text documents, images, audio, mind maps or videos. Often the documents themselves are not kept or stored directly in the IR system, but are instead represented in the system by document surrogates or metadata. Most IR systems compute a numeric score on how well each object in the database matches the query, and rank the objects according to this value. The top ranking objects are then shown to the user. The process may then be iterated if the user wishes to refine the query.

2.

First dimension: mathematical basis Set-theoretic models represent documents as sets of words or phrases. Similarities are usually derived from set-theoretic operations on those sets. Common models are: Standard Boolean model Extended Boolean model Fuzzy retrieval Algebraic models represent documents and queries usually as vectors, matrices, or tuples. The similarity of the query vector and document vector is represented as a scalar value. Vector space model Generalized vector space model (Enhanced) Topic-based Vector Space Model Extended Boolean model Latent semantic indexing a.k.a. latent semantic analysis Probabilistic models treat the process of document retrieval as a probabilistic inference. Similarities are computed as probabilities that a document is relevant for a given query. Probabilistic theorems like Bayes' theorem are often used in these models. Binary Independence Model Probabilistic relevance model on which is based the okapi (BM25) relevance function Uncertain inference Language models Divergence-from-randomness model Latent Dirichlet allocation Feature-based retrieval models view documents as vectors of values of feature functions (or just features) and seek the best way to combine these features into a single relevance score, typically by learning to rank methods. Feature functions are arbitrary functions of document and query, and as such can easily incorporate almost any other retrieval model as just another feature. Data fusion models: Data fusion in information retrieval combines results from multiple search systems or retrieval models to improve performance. By merging ranked lists, it leverages the strengths of diverse approaches, often enhancing recall and precision. Common methods include score normalization and voting techniques like CombSUM or Borda count. This meta-search strategy is particularly effective when individual systems have complementary coverage or when query difficulty varies, producing a more robust and reliable final ranking sparsity.

3.

History there is ... a machine called the Univac ... whereby letters and figures are coded as a pattern of magnetic spots on a long steel tape. By this means the text of a document, preceded by its subject code symbol, can be recorded ... the machine ... automatically selects and types out those references which have been coded in any desired way at a rate of 120 words a minute. The idea of using computers to search for relevant pieces of information was popularized in the article *As We May Think* by Vannevar Bush in 1945. It would appear that Bush was inspired by patents for a 'statistical machine' – filed by Emanuel Goldberg in the 1920s and 1930s – that searched for documents stored on film. The first description of a computer searching for information was described by Holmstrom in 1948, detailing an early mention of the Univac computer. Automated information retrieval systems were introduced in the 1950s: one even featured in the 1957 romantic comedy *Desk Set*. In the 1960s, the first large information retrieval research group was formed by Gerard Salton at Cornell. By the 1970s several different retrieval techniques had been shown to perform well on small text corpora such as the Cranfield collection (several thousand documents). Large-scale retrieval systems, such as the Lockheed Dialog system, came into use early in the 1970s. In 1992, the US Department of Defense along with the National Institute of Standards and Technology (NIST), cosponsored the Text Retrieval Conference (TREC) as part of the TIPSTER text program. The aim of this was to look into the information retrieval community by supplying the infrastructure that was needed for evaluation of text retrieval methodologies on a very large text collection. This catalyzed research on methods that scale to huge corpora. The introduction of web search engines has boosted the need for very large scale retrieval systems even further. By the late 1990s, the rise of the World Wide Web fundamentally transformed information retrieval. While early search engines such as AltaVista (1995) and Yahoo! (1994) offered keyword-based retrieval, they were limited in scale and ranking refinement. The breakthrough came in 1998 with the founding of Google, which introduced the PageRank algorithm, using the web's hyperlink structure to assess page importance and improve relevance ranking. During the 2000s, web search systems evolved rapidly with the integration of machine learning techniques. These systems began to incorporate user behavior data (e.g., click-through logs), query reformulation, and content-based signals to improve search accuracy and personalization. In 2009, Microsoft launched Bing, introducing features that would later incorporate semantic web technologies through the development of its Satori knowledge base. Academic analysis have highlighted Bing's semantic capabilities, including structured data use and entity recognition, as part of a broader industry shift toward improving search relevance and understanding user intent through natural language processing. A major leap occurred in 2018, when Google deployed BERT (Bidirectional Encoder Representations from Transformers) to better understand the contextual meaning of queries and documents. This marked one of the first times deep neural language models were used at scale in real-world retrieval systems. BERT's bidirectional training enabled a more refined comprehension of word relationships in context, improving the handling of natural language queries. Because of its success, transformer-based models gained traction in academic research and commercial search applications. Simultaneously, the research community began exploring neural ranking models that outperformed traditional lexical-based methods. Long-standing benchmarks such as the Text REtrieval Conference (TREC), initiated in 1992, and more recent evaluation frameworks Microsoft MARCO (MACHINE Reading COMprehension) (2019) became central to training and evaluating retrieval systems across multiple tasks and domains. MS MARCO has also been adopted in the TREC Deep Learning Tracks, where it serves as a core dataset for evaluating advances in neural ranking models within a standardized benchmarking environment. As deep learning became integral to information retrieval systems, researchers began to categorize neural approaches into three broad classes: sparse, dense, and hybrid models. Sparse

models, including traditional term-based methods and learned variants like SPLADE, rely on interpretable representations and inverted indexes to enable efficient exact term matching with added semantic signals. Dense models, such as dual-encoder architectures like ColBERT, use continuous vector embeddings to support semantic similarity beyond keyword overlap. Hybrid models aim to combine the advantages of both, balancing the lexical (token) precision of sparse methods with the semantic depth of dense models. This way of categorizing models balances scalability, relevance, and efficiency in retrieval systems. As IR systems increasingly rely on deep learning, concerns around bias, fairness, and explainability have also come to the picture. Research is now focused not just on relevance and efficiency, but on transparency, accountability, and user trust in retrieval algorithms.